

Wave Basics + Standard Form

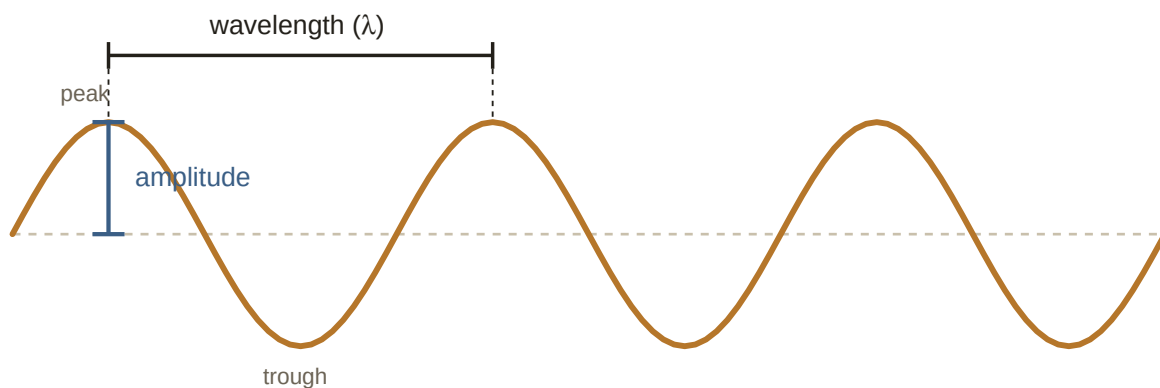
Quick reference sheet · Physics & Maths · keep handy for revision

1. Two kinds of wave

A wave transfers **energy** from one place to another without the material itself travelling along with it.

- **Transverse waves** — particles move *at right angles* to the direction of travel. Examples: light, water waves, a wiggled rope.
- **Longitudinal waves** — particles move *back and forth along the same line* as the wave travels, creating compressions and rarefactions. Examples: sound, a pushed slinky.

2. Reading a wave



- **Amplitude** — maximum displacement from the rest position. Bigger amplitude = more energy.
- **Wavelength (λ)** — distance between two identical points on the wave (e.g. peak to peak). Measured in metres.
- **Frequency (f)** — number of complete waves passing a point per second. Measured in hertz (Hz).
- **Wave speed equation:** $v = f \times \lambda$ (speed in m/s = frequency in Hz $\times$ wavelength in m)

3. Standard form

Standard form writes any number as $a \times 10^n$, where a is between 1 and 10, and n is a whole number (positive for big numbers, negative for small ones).

Example: 0.0000005 m $\rightarrow$ move the decimal point 7 places $\rightarrow 5 \times 10^{-7}$ m

Wave type	Ordinary number	Standard form
Radio wave	1500 m	1.5×10^3 m
Visible light	0.0000005 m	5×10^{-7} m
X-ray	0.0000000001 m	1×10^{-10} m

Tip: the number of places you move the decimal point is the power of 10. Moving left on a big number gives a positive power; moving right on a small decimal gives a negative power.